

House Price Prediction

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Project Summary

The aim of this project was to build a machine learning model that could predict house prices using various property-related features such as area, number of bedrooms, bathrooms, parking spaces, stories, furnishing status, and amenities like air conditioning and guest rooms. The project involved data preprocessing, exploratory data analysis, model training, evaluation, and interpretation of results.

The dataset was first cleaned and prepared for analysis. Since several features were categorical in nature (such as yes/no attributes and furnishing status), they were converted into numerical values using one-hot encoding. The data was then divided into training and testing sets to ensure that the models could be evaluated on unseen data.

Two regression algorithms were used for prediction: Linear Regression and Random Forest Regressor. After evaluating both models using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score, Linear Regression emerged as the better-performing model. It achieved an R^2 score of approximately 0.653, meaning that the model was able to explain around 65% of the variation in house prices. The Random Forest model achieved a slightly lower R^2 score of approximately 0.613. This result indicates that the **relationship between the housing features and price is relatively linear, allowing the simpler Linear Regression model to outperform the more complex Random Forest model**.

The histogram of house prices revealed that the dataset follows a right-skewed distribution. **Most properties are concentrated in the lower-to-middle price range**, while only a small number of houses fall into the premium price category. This pattern is common in real estate markets where affordable and mid-range properties are more frequent than luxury homes.

The correlation heatmap provided valuable insights into the relationships between variables. Among all features, **area showed the strongest positive correlation with house price**, indicating that larger properties tend to be significantly more expensive. Bathrooms, stories, parking spaces, and air conditioning also demonstrated moderate positive correlations with

price. On the other hand, unfurnished houses showed a negative relationship with price, suggesting that furnished or semi-furnished properties generally command higher market values.

The Actual vs Predicted Price scatter plot showed a **clear positive trend between actual house prices and the model's predictions**. Most points follow an upward pattern, indicating that the model successfully captures the general pricing trend. However, some dispersion can be observed for higher-priced properties, suggesting that the model finds it more challenging to predict extremely expensive houses with complete accuracy. This is expected because luxury properties often depend on additional factors not included in the dataset.

Feature importance analysis further confirmed that **area was the most influential factor affecting house prices, contributing nearly 47%** of the total importance. Bathrooms were the second most important feature, followed by air conditioning availability, parking capacity, number of stories, and bedrooms. These findings highlight that **both property size and lifestyle-related amenities play a major role in determining market value**.

Overall, the project successfully demonstrated how machine learning can be used to estimate house prices based on property characteristics. The results show that larger houses with more bathrooms, parking spaces, and modern amenities tend to have higher prices. One interesting observation was that the simpler Linear Regression model performed better than the Random Forest model, which was unexpected but indicates that the dataset follows relatively straightforward patterns. Based on these findings, real estate businesses should **focus on properties with larger living spaces and premium amenities**, as these factors have the strongest influence on pricing and buyer demand.

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